

Qatar-1b

R-2796

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_120819 · created 2026-09-04T23:12:59+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456158.84157 +/- 0.00015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.166 +/- 0.047
Transit depth (Rp/Rs)^2	2.76 +/- 1.55 [%]
Orbital Inclination (inc)	84.76 +/- 2.29
Airmass coefficient 1 (a1)	1.052 +/- 0.04
Airmass coefficient 2 (a2)	-0.028 +/- 0.04
Scatter in the residuals of the lightcurve fit is	3.8 %
Best Comparison Star	#2 - [286, 286]
Optimal Aperture	2.48
Optimal Annulus	7.06
Transit Duration (day)	0.07 +/- 0.014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.166 ± 0.047 vs 0.1463 (deviation 0.42σ)
Duration fit vs published	0.07 d vs 0.0692 d (deviation 0.06σ)
Mid-time O–C	-0.4 min
Depth SNR	3.5
Residual scatter	3.8 %

Light curve

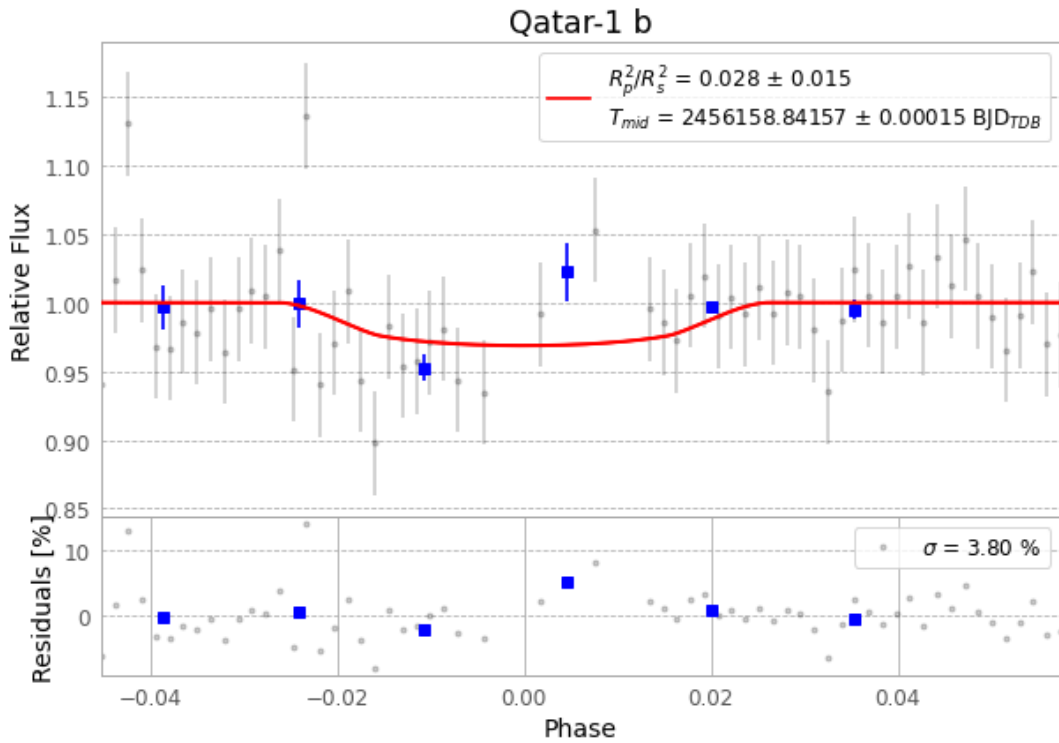


Figure 1 — FinalLightCurve_Qatar-1 b_2012-08-18.png (SHA-256 799c88ecf8a5de73...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [355, 263], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9019f6978952fbbb...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

71 FITS frames (47 MB), Qatar-1120819063612.FITS → Qatar-1120819100612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-120819.log (36f5ffad249c...), FinalLightCurve_Qatar-1 b_2012-08-18.png (799c88ecf8a5...), FinalLightCurve_Qatar-1 b_2012-08-18.csv (7ca53f34a4d5...)

Compute resources

Wall time 395.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-04 23:13Z.